



2022 CHINSAPO CLUSTER MOCK

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION

BIOLOGY

Date___/___/ 2022

PAPER I (100 marks)

Subject Number: **MO22/1**

Time allowed: 2hours

INSTRUCTIONS

1. This paper contains 12 printed pages please check
2. Before you begin, fill in your name and class at the top of the question paper and on all other sheets
3. This paper contains two sections A and B. Answer all questions in all the sections. some can be answered quickly but others will require considerable
4. Write your answers on the question paper in the spaces provided. The maximum number of marks for each answer is indicated against each question
5. In the table provided on this page tick against the question number you have answered
6. Hand in your question paper to the invigilator when time is called to stop writing

Question number	Tick if answered	Do not write in these columns	
1			
2			
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TOTAL			

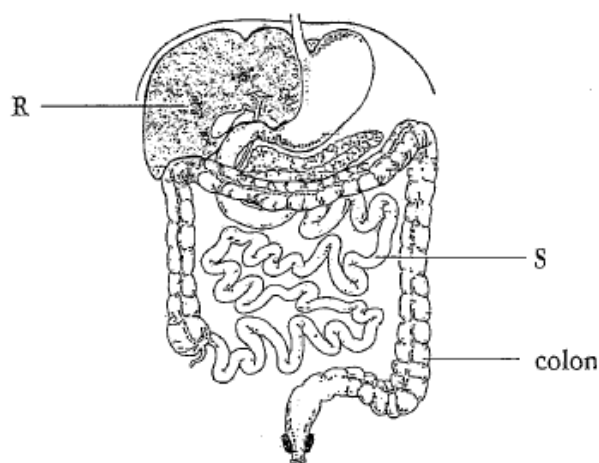
1. a. Differentiate Hypertonic from Hypotonic solutions?

_____2marks

- b. By what process does a nitrate ion move into a root hair cell when its concentration is higher in the root cell than in the soil.

_____1mark

2. **Figure 1** is a diagram showing parts of the human digestive system .Use it to answer the questions.



- a) Name the parts marked R and S

R _____ (1mark)

S _____ (1 mark)

- b) Explain how part R ensures a steady supply of glucose to the body

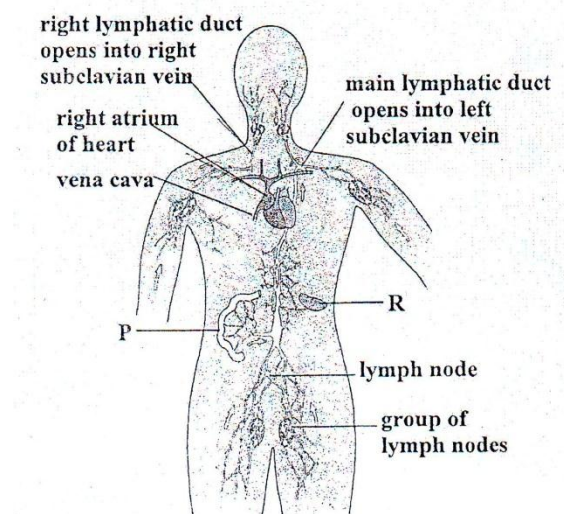
- c) If the part R has failed to produce bile, what two effects could this have on digestion

_____ (2marks)

- d) Explain why the level of insulin increase in the blood soon after a meal of carbohydrates

_____2marks

3. . **Figure 2** is a diagram showing the lymphatic system. Use it to answer the questions that follow.



a) Name the part labelled **R**

_____ (1 mark)

b) i. Define Lymph

_____ 1mark

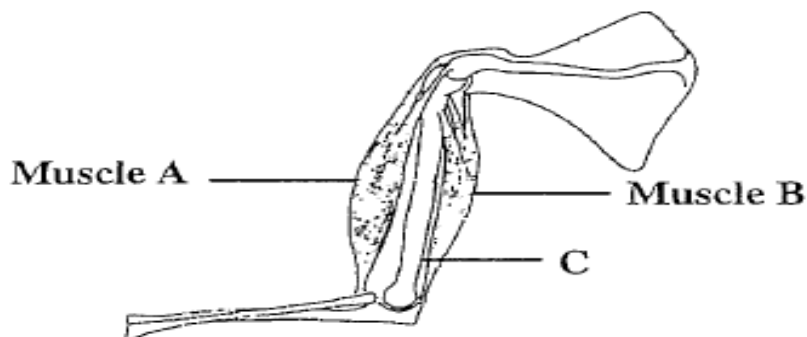
ii .State any one difference between Lymph and Plasma

_____ 1mark

iii. Explain how inspirational movement of the chest wall helps in the flow of Lymph

_____ 3marks

4. **Figure 3** is a diagram showing antagonistic muscles of the arm



CANDIDATE NAME _____

a) i. What is the name of muscle A

_____ 1mark

ii. Name the part marked C

_____ 1mark

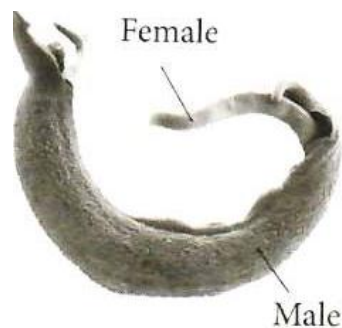
b) How does contraction of muscle B affect the arm

_____ 1mark

c) State two injuries which may occur at a joint

_____ 2marks

5. **Figure 4** is a diagram showing two parasitic worms .Use it to answer questions 8 to 10.



a) Identify the parasites

_____ 1mark

b) Name the disease caused by these parasites

_____ 1mark

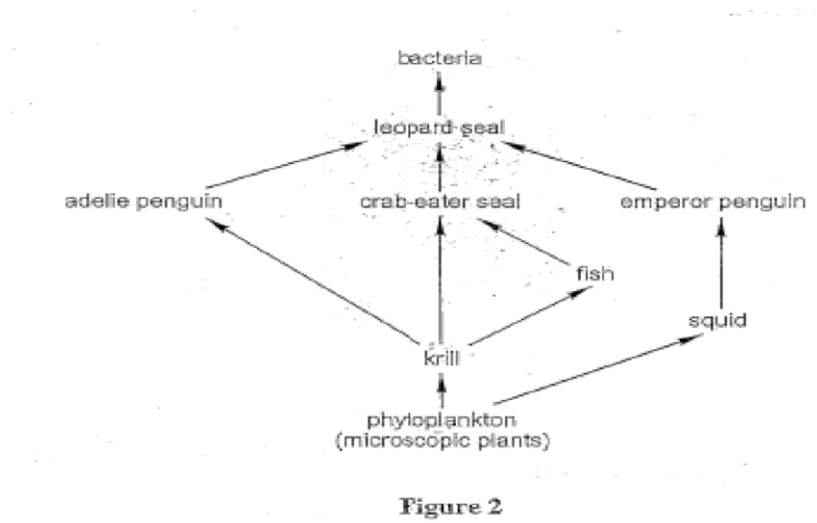
c) State any one mode of transmission of the disease in 5b

_____ 1mark

d) Give any one way in which the disease can be controlled

_____ 1mark

6. The **figure 5** below is a diagram showing a food web in an aquatic ecosystem. Use it to answer the questions that follows



- a) Name one organism which represents
- Herbivore _____ 1mark
 - Decomposer _____ 1mark

- b) .From the food web,draw one food chain of six organisms

_____ 2marks

- c) Why would bacteria not belong to specific feeding level

_____ 1mark

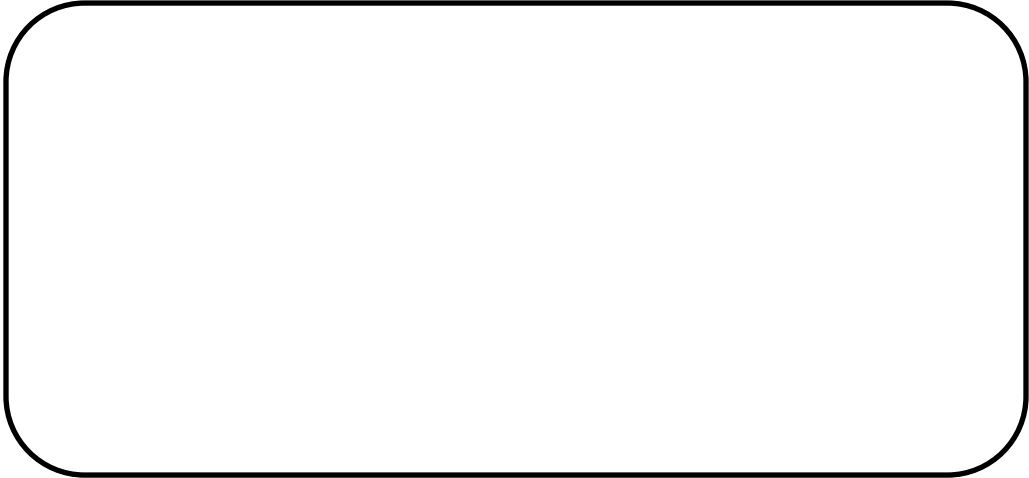
- d) Give two ways in which the activities of man can affect an aquatic ecosystem

 _____ 4marks

CANDIDATE NAME _____

7. When a grey cock was mated with a grey hen, grey and white chicks were produced.

- a) Using G for grey colour and g for white colour, draw a genetic diagram to determine the genotype of the offspring



3marks

- b) Give the genotypic ratio of the chicks

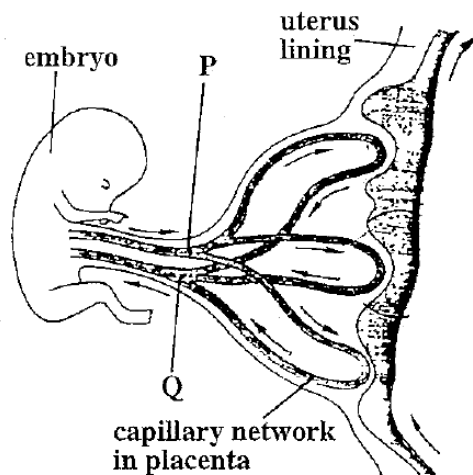
1mark

- c) If the parents produced 12 chicks, how many were white, show your working



3marks

8. The **figure 6** below is a diagram showing blood supply between an embryo and the placenta. Use it to answer the questions that follow



- a) Name the blood vessels marked P and Q

- i. P _____ 1mark
 ii. Q _____ 1mark

- b) Mention any two substances transported by blood vessel marked P

- i. _____ 1 mark
 ii. _____ 1 mark

- c) Name one organ that starts to function immediately a child is born

_____ 1mark

- d) Explain one adaptation of the placenta to its function

_____ 2marks

9. (a). Define auxins

 _____ (1 Mark)

- a) . How does high auxin concentration affect the growth of the following parts of the plants?

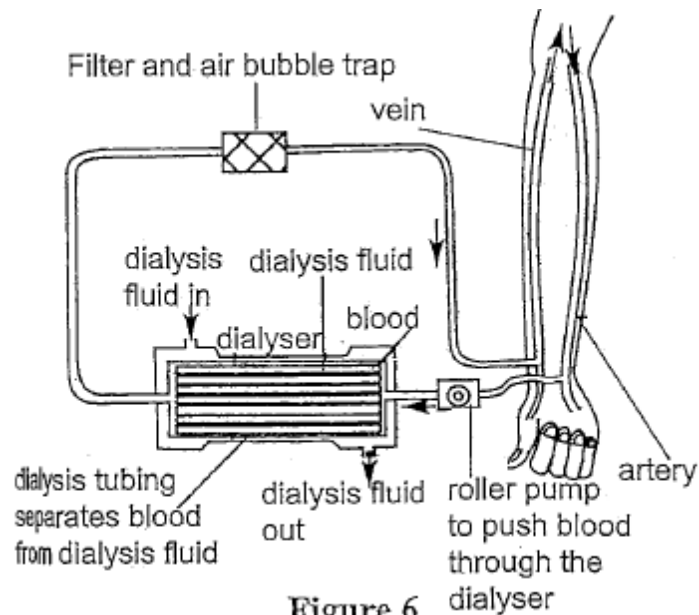
- i) Shoots

_____ (1 Mark)

- i) Roots

_____ (1 Mark)

10. The **figure 7** below shows a dialysis machine. Use it to answer the questions that follows



a) Name two substances which diffuse out of the dialysis tube when it is in operation

_____2marks

b) Explain why patient's blood and the dialysis fluid move to opposite directions in the dialyser

_____2marks

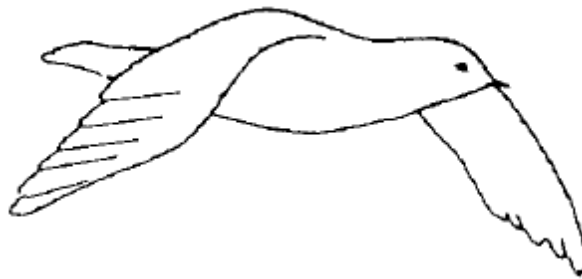
c) Why are there many smaller channels in the dialysis rather than one larger one

_____2marks

d) Explain why it is dangerous for an air bubble to get into the patients blood while on the dialysis machine

_____2marks

11. The **figure 8** below shows a bird in flight .Use it to answer the questions that follows



a) Identify the stroke shown in the bird

_____1mark

b) Explain how the stroke shown in the stroke named in (a) occurs

_____2marks

12. The figure below shows a mode of transmission of some diseases. Use it to answer the questions that follows



a) .Identify the mode of transmission

_____1mark

b) Name any two diseases that can be transmitted by the mode shown in the figure above

_____2marks

c) Explain one way of preventing the transmission of the disease through the mode shown in the figure

_____2marks

CANDIDATE NAME _____

SECTION B ESSAY QUESTIONS 30 MARKS

13. Discuss five adaptations of Hydrophytes in an essay form

_10marks

CANDIDATE NAME _____

14. Describe the steps that could be followed in artificial resuscitation to make a drowned person regain breathing

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general note-taking. There are no margins, text, or other markings on the page.

10marks

